The confinement scale in three bands, with published numbers, and the verification restated gauge-invariantly

Later qualifications: reasons to stop rejects the proposed crossover certificate as a currently available route; mass-gap openings §1 warns of scalar softening; obstruction map identifies the unsupported blocking and continuum claims.

> **Correction (2026-10-02).** The earlier claim that three strong-side blocking steps are standard once mixing is certified is withdrawn. Gibbsianness does not bound the generated interaction, preserve Wilson form, or establish mixing at successive scales. The identification of a coarse Wilson-loop expectation with polymer activity is heuristic; the unsourced activity threshold 2×10^{-3} is withdrawn. So are the claim that six steps constitute the whole continuum problem, the comparison making band B more strongly mixing than band A, and the unsourced series singularities at $\beta_W = 3.5$–4. The table below gives continuum-scaled estimates, not measured lattice masses or uniform boundary-mixing bounds. Box sides 3–5 are trial sizes, not certified sufficient sizes; σa^2 is an area coefficient, not a boundary-influence decay rate. The physical energy conversion is corrected to $8.42\,\hbar c\,\mathrm{fm}^{-1} \simeq 1.66$ GeV. The original 216–2000 link count is retained with its explicit four-dimensional, free-boundary convention; the earlier description as merely a few hundred links was inconsistent with that convention.

This note gives an exact gauge-symmetry observation and a numerical orientation map for pure $SU(3)$ Wilson theory in four Euclidean dimensions. It corrects the proposed single-link block test in the Dobrushin note §4 and the finite-verification note. The scale estimates motivate possible sub-block mixing tests; they supply neither such a certificate nor a quantitative blocking bridge nor a continuum existence or mass-gap theorem.

Elitzur empties the interior single-link test. Let V be a set of links and s a site all of whose incident links lie in V. A gauge transformation at s preserves the Wilson conditional Gibbs measure μ_V^ω for every fixed external boundary condition ω. It acts on any incident link by arbitrary left or right multiplication in $SU(3)$. Its marginal is consequently Haar, independently of ω. Thus the single-link boundary influence $\rho_V(x, y)$ vanishes for every link x with such an endpoint. A sum of these single-link influences sees only the remaining boundary layer. This observation alone proves no uniqueness or decay of joint interior observables: identical one-link marginals need not give identical joint distributions.

A useful **target** is instead a sub-block estimate, with d measured in the link interaction graph,

$$\left\| \mu_V^\omega\big|_W - \mu_V^{\omega'}\big|_W \right\|_{\mathrm{TV}} \;\le\; C\,|W|\,e^{-\gamma d(W,y)}, \qquad W \subseteq V,$$

for boundary conditions differing at the external link y. Gauge symmetry removes the orbit variables at sites whose incident links all lie in W. Closed-loop observables then provide non-trivial tests of the invariant content. A selected set of small Wilson-loop traces is not asserted to determine the full marginal; boundary-attached observables and the full joint invariant distribution may also be needed. The single-link Dobrushin test of the Dobrushin note §2 is unaffected: conditioning on all other links removes the free gauge transformation used above.

The displayed inequality is a target decay form, not by itself a finite-size certificate. On one fixed box an arbitrarily large C can make it vacuous. A theorem would require an applicable quantitative finite-size criterion, a verified contraction margin and control of the resulting constants across volumes. No equivalence to complete analyticity for this gauge specification is proved here.

Published scales give orientation, not certified box sizes. Use r_0/a from Necco and Sommer (Nucl. Phys. B 622 (2002) 328; metadata level) and the continuum estimate

$r_0\mu_{0^{++}} = 4.21$ from Morningstar and Peardon (Phys. Rev. D 60 (1999) 034509; abstract read, numerical input retained at metadata level). Here $\mu_{0^{++}}$ is an inverse length and the rest energy is $E_{0^{++}} = \hbar c \mu_{0^{++}}$. The continuum-scaled columns use

$$\mu_{0^{++}} a = \frac{4.21}{r_0/a}, \qquad \frac{\xi}{a} = \frac{r_0/a}{4.21}, \qquad \sigma a^2 = \frac{1.35}{(r_0/a)^2},$$

where $r_0^2\sigma \simeq 1.35$ is an orienting string-tension input; σ has units of inverse area, with physical tension $\hbar c \sigma$.

β_W	$g^2 = 6/\beta_W$	r_0/a	scaled $\mu_{0^{++}} a$	scaled ξ/a	scaled σa^2
5.7	1.05	2.92	1.44	0.70	0.16
6.0	1.00	5.37	0.78	1.28	0.047
6.2	0.97	7.38	0.57	1.75	0.025
6.4	0.94	9.74	0.43	2.31	0.014

Caveat, 2026-09-23, retained. Near $\beta_W \simeq 5.7$ the Wilson axis passes beside the endpoint of the fundamental–adjoint first-order line, where the scalar glueball softens (Heller 1995; metadata level, openings note §1). The measured lattice ξ/a can therefore exceed the scaled value 0.70. Even a measured vacuum correlation length would not certify mixing uniformly over all boundary conditions.

A trial box with $R = 3$–5 **sites per axis in four dimensions**, with only internal nearest-neighbour links integrated, has

$$|V| = 4(R-1)R^3 = 216,\ 768,\ 2000 \qquad (R = 3, 4, 5).$$

There are R^3 choices of transverse coordinates and $R-1$ links in each of four directions. This is hundreds to two thousand links; three-dimensional spatial slices have a different count. Neither the scalar estimate nor $\sigma a^2 \simeq 0.16$ fixes a sufficient R. An area law concerns large loop areas. Temporal decay in a given channel is governed by its lowest energy with nonzero overlap; a flux-tube energy depends on its length and receives corrections. Replacing any of these by a boundary-influence rate σa^2 is an unproved heuristic.

1. Three bands of the coupling

The boundaries below label the original comparison, not proven phase boundaries or the optimal reach of every method.

band	β_W	status
A	< 0.0135	single-link Dobrushin condition holds; fixed-cutoff Wilson transfer gap in the local-observable sector, uniformly in volume
B	0.0135 to about 5.7	outside that sufficient condition; confinement is physically expected, but no mixing certificate is supplied here

band	β_W	status
C	above about 5.7	scaled ξ/a grows in the table; weak-coupling RG control toward the continuum remains an open obligation

For band A the exact sufficient threshold from the one-link estimate is $\beta_W < \frac{1}{4}\log(19/18)$. The convenient $g^2 > 444$ implies it: $\log(19/18) = \int_{18}^{19} dt/t > 1/(18.5) = 2/37 = 24/444$, by strict convexity of $1/t$ and the midpoint integral bound. The normalized positive Wilson transfer operator $\mathcal{T}$ then has an energy-gap bound on the cyclic local-observable sector

$$\Delta_W \geq \frac{\hbar c}{a} \log \frac{1}{18(e^{24/g^2} - 1)} > 0,$$

as explained in the Dobrushin note §§2–3 (written derivation read). The volume uniformity is at fixed a, g^2; this is not a continuum limit.

Band B is a proposed sharpness problem. The crude sufficient criterion leaves a wide interval. It does not follow that band B is more mixing than band A: as $\beta_W \to 0$ the measure approaches a product of independent Haar links, and the Dobrushin decay bound improves without limit. Nor does the table establish $\xi/a < 1$ throughout band B. The all-representation activity estimate in the Wilson note is of order β_W with a quoted sufficient activity scale of order 0.0057; the leading fundamental activity is a different normalization. Changing surface entropy counts alone is not a proved extension to $\beta_W \sim 0.1$. The earlier assertion of series singularities at 3.5–4 is withdrawn for lack of a source or specified series.

Band C is an RG target. For orientation only, the one-loop pure-$SU(3)$ running has $b_0 = 11/(16\pi^2)$ and

$$\frac{1}{g^2(2a)} - \frac{1}{g^2(a)} = -2b_0 \log 2 \simeq -0.0966.$$

Going from $g^2 = 1/2$ to $g^2 = 1$ therefore takes $1/0.0966 \simeq 10.4$ doublings in this approximation. At fixed physical correlation length, a decrease of ξ/a from 10 to 1 takes $\log_2 10 \simeq 3.32$ doublings kinematically. Neither count establishes constructive control, locates gap formation, or proves a gap of order $\hbar c/a$ throughout band B.

2. What the verification costs, restated

For the trial boxes just defined, the un-gauge-fixed bulk integral has $8|V| = 1728, 6144, 16000$ real dimensions, since $\dim SU(3) = 8$. External links entering plaquettes meeting V give an additional boundary supremum over $8|\partial V|$ variables, once that boundary link set is specified. Gauge reduction may decrease the number of independent coordinates, but does not provide a certified marginal bound.

A verification would have to control joint sub-block distributions uniformly over those boundary conditions, satisfy a quantitative finite-size criterion with margin, and state its channel and decay rate. No such verification is performed here. The proposed small boxes are exploratory sizes; the earlier claim of a three-to-four order reduction in the required problem size is withdrawn because no sufficient size is known. The later reasons-to-stop note (relevant passages read) rejects physical-crossover certification as a currently supplied route and redirects effort toward weak-side RG. Its historical box counts do not replace the explicit four-dimensional count above.

2b. A heuristic meeting scale, not a six-step theorem

From the weak side. The one-loop count suggests about ten doublings from $g^2 = 1/2$ to $g^2 \simeq 1$. It does not establish that the first seven steps are controlled or that only three steps are open.

Uniform fluctuation bounds, generated-interaction bounds and iteration remain required even in the weak-coupling part of the proposed trajectory.

From the strong side: the area-law illustration. Choose coarse links as ordered products of straight fine links. Then a coarse plaquette is exactly the holonomy of a $2^k \times 2^k$ fine loop after k doublings. If one models its normalized expectation by the pure area term $e^{-\sigma a^2 4^k}$ with $\sigma a^2 = 0.16$, the estimates are 0.53, 0.077, 3.6×10^{-5} for $k = 1, 2, 3$. These are values of that heuristic model: perimeter terms, prefactors, finite-loop effects and lattice artifacts are omitted. In particular, the asymptotic area law need not approximate a 2×2 loop.

An expectation is not a polymer activity or a bound on the norm of an interaction. The blocked measure generally contains interactions beyond Wilson plaquettes. The former 2×10^{-3} threshold had no specified activity norm or source and is withdrawn; no conclusion that three doublings reach band A follows. A Migdal–Kadanoff fourth-power recursion (Kadanoff, Ann. Phys. 100 (1976) 359; metadata level) is likewise a heuristic in four dimensions and supplies no certified step count or error comparison here.

A measurable block map has a well-defined pushforward probability measure. Representing it by a summable Gibbs interaction requires additional hypotheses on that map and measure. The abstract of van Enter, Fernández and Sokal (J. Stat. Phys. 72 (1993) 879; abstract read) explicitly distinguishes regularity on the RG map's domain from examples of non-Gibbsian images. This reference is not a quantitative theorem for the present $SU(3)$ map. Even if a suitable Gibbs representation were established, three steps would still require explicit bounds on every generated interaction, stability or renewed mixing estimates, and a justified comparison with a strong-coupling criterion. None is supplied here.

The meeting point is a proposed target. A reference interaction near $\beta_W = 5.7$–6 is suggested by the numerical scales. A certified box there would concern that interaction and its proven stability neighbourhood; it would not automatically certify all smaller β_W or all its blocked images. The earlier identifications $g^2_{\mathrm{DS}} \simeq 1$ and $g^2_{\mathrm{RG}} \geq 1$ are conjectural targets, not thresholds established by this note.

The former "whole problem in one sentence" claim is withdrawn. The map motivates studying a few physical scales around order-one coupling, but does not reduce the continuum mass-gap problem to six blocking steps. The conditional-theorem note organizes proposed mixing and RG hypotheses; invoking it here does not discharge them. Continuum existence, Osterwalder–Schrader or Wightman axioms, nontriviality, and a finite positive physical spectral gap remain separate obligations, as the spectral obligations require.

3. What this changes

The exact correction is the Haar-marginal obstruction to an interior single-link test. A joint sub-block criterion is a research target; its constants, applicability and sufficient box size remain open. The three bands and blocking counts are orientation, not an established connection between the explicit strong-coupling region and the physical crossover.

For the conventional scale choice $r_0 = 0.5$ fm, the retained continuum ratio gives

$$\mu_{0^{++}} = \frac{4.21}{r_0} = 8.42\,\mathrm{fm}^{-1}, \qquad E_{0^{++}} = 8.42\,\hbar c\,\mathrm{fm}^{-1} \simeq 1.66\,\mathrm{GeV},$$

using $\hbar c \simeq 0.1973$ GeV fm. If mass is expressed as a mass rather than an energy, $M_{0^{++}} = E_{0^{++}}/c^2 \simeq 1.66$ GeV$/c^2$. This is a lattice-literature benchmark with a chosen physical scale, not a proved continuum gap or a requirement that a lower-bound proof recover that central numerical value.

4. Consequence for STATE

Retain the exact interior Haar-marginal obstruction and the explicit fixed-cutoff strong-coupling criterion. Treat the scale table, $R = 3$–5 trial boxes, area-law blocking illustration and proposed

crossover meeting point as orientation only. Withdraw the claims that one such box certifies band B, that three subsequent strong-side steps are standard, or that six steps settle the continuum theory. The later reasons-to-stop correction points toward quantitative weak-side RG and control of generated interactions; this note supplies no new certificate, blocking theorem or continuum construction.